MONASH University

AgriSuit

by

MATRIX GEOSCIENCE

Azham Hassan V (34484841)
Rafli Ramdani (34347356)

Agenda

- Problem Statement
- Motivation
- Methodology
- Key Features & Demonstration
- Future Scope & Conclusion

Matrix Geoscience is an Australian consultancy delivering practical, mathematical solutions to streamline operations.

- Serves mining, geology, education, materials handling. worldwide
- Geo-statistics, data analytics, and ML expertise.
- Tailored, efficiency-focused implementations.

What pressing challenge does AgriSuit aim to solve?

- Farmers and investors in Tasmania find it hard to choose land that truly fits their farming goals.
- Real-estate sites show what's for sale, not what's suitable.
- They don't include key details like soil, climate, or zoning.

This leads to **missed opportunities** and slow, manual land assessment.

Motivation

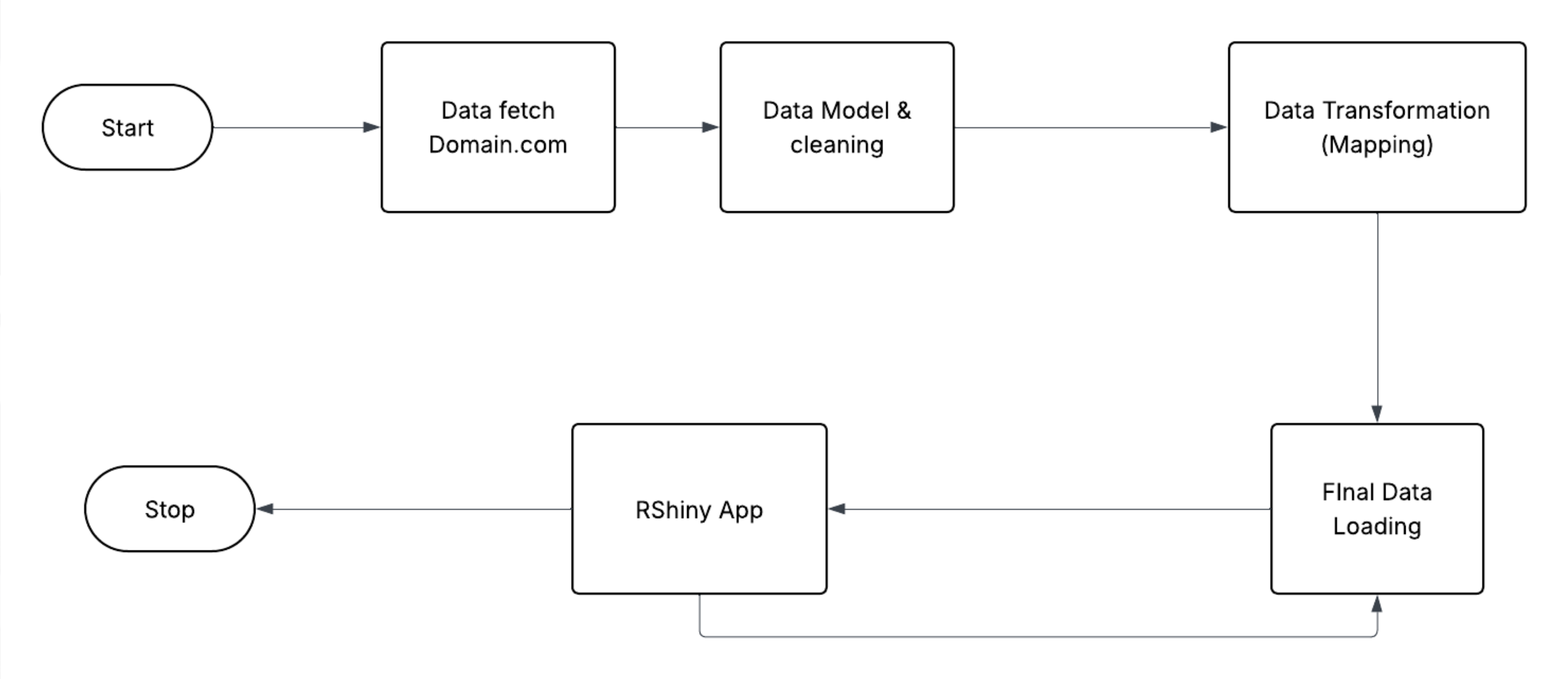
- **Agriculture contributes ~7–9% to Tasmania's GSP**, driving regional growth *(Department of Treasury and Finance Tasmania, 2024)*.
- TESM (Tasmanian Enterprise Suitability Map) not linked to live property markets.
- **AgriSuit** bridges that gap, combining TESM data with property listings.
- Enables data-driven and climate-resilient land investment.

Methodology

Data Sources

- **Domain.com.au** for *property listings data*.
- **LISTMap open data** in for the Statewide *Cadastral Parcel*, *Address Points*, and *Crop Rules / Suitability Map*.
- Basemaps (Web-hosted Maps): **CartoDB Positron**; **LIST Topographic, LIST Orthophoto.**

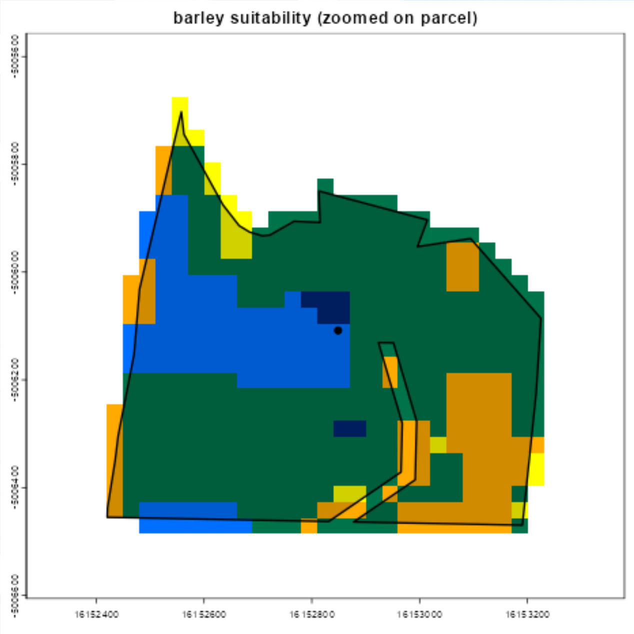
Project Flow



Final Table

FULL_ADDRESS	MUNICIPALITY	CROP	CLASS_LABEL	PRICE (AUD)	COUNT	WEBSITE
120 Russells Plains Road Rocherlea TAS 7248 Australia	LAUNCESTON	Barley	1.1 Well suited (with soil management)	550,000	31	View
120 Russells Plains Road Rocherlea TAS 7248 Australia	LAUNCESTON	Barley	4.0 Unsuitable	550,000	239	View
120 Russells Plains Road Rocherlea TAS 7248 Australia	LAUNCESTON	Barley	2.0 Suitable	550,000	105	View
167 Adams Road Lebrina TAS 7254 Australia	LAUNCESTON	Barley	1.1 Well suited (with soil management)	3,000,000	20	View
167 Adams Road Lebrina TAS 7254 Australia	LAUNCESTON	Barley	2.1 Suitable (with soil management)	3,000,000	62	View

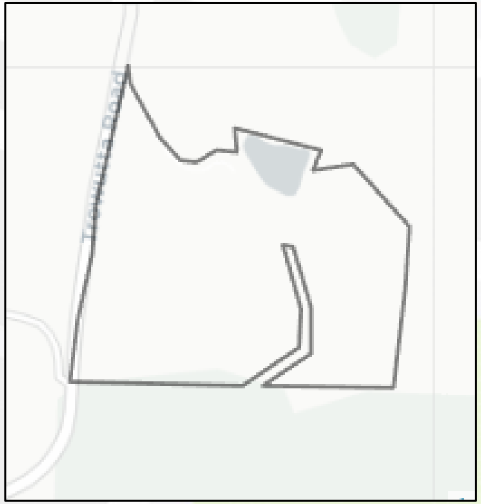
Pixel Mapping



ESM suitability classes

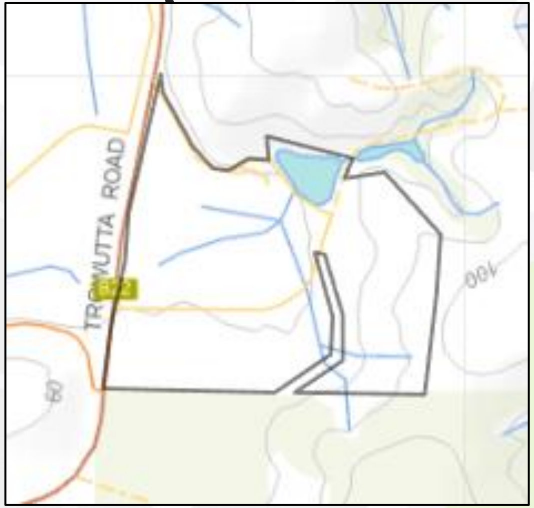
- 1.0 Well suited
- 1.1 Well suited (with soil management)
- 2.0 Suitable
- 2.1 Suitable (with soil management)
- 3.0 Moderately suitable
- 3.1 Moderately suitable (with soil management)
- 4.0 Unsuitable

Web Map Service Mapping



CartoDB Map

LIST Satelite



LIST Topography

Key Features

- Interactive map view showing parcel-level crop suitability.
- Smart filters: crop, class, municipality, and price range.
- Direct property listing links with suitability data.
- User registration for preference saving and alerts.
- Combined map + table results for complete insights.

Demonstration

Future Scope

- Add 2030 & 2050 climate projections.
- Enable auto database refresh and email alerts.
- Expand to mainland Australia.
- Personalised E-mail based Notification

Commercialisation

- Replace scraping with API integrations for agricultural land listings
- Move to cloud-based backend for scalability
- Enable automated daily data refreshes
- Add premium features (ROI analysis, alerts, crop comparison)
- Transition toward a fully commercial, investor-ready platform

Conclusion

- **AgriSuit** bridges the gap between geospatial intelligence and the property market.
- Empowers investors and farmers with science-backed land insights.
- Demonstrates how open data can drive sustainable growth.
- “**AgriSuit**, Where Every Land Finds Its Purpose.”

Thanks!

Do you have any questions?